OPERATING SYSTEMS AND COMPUTER NETWORKS

FUNDAMENTALS OF NETWORK PROTOCOLS

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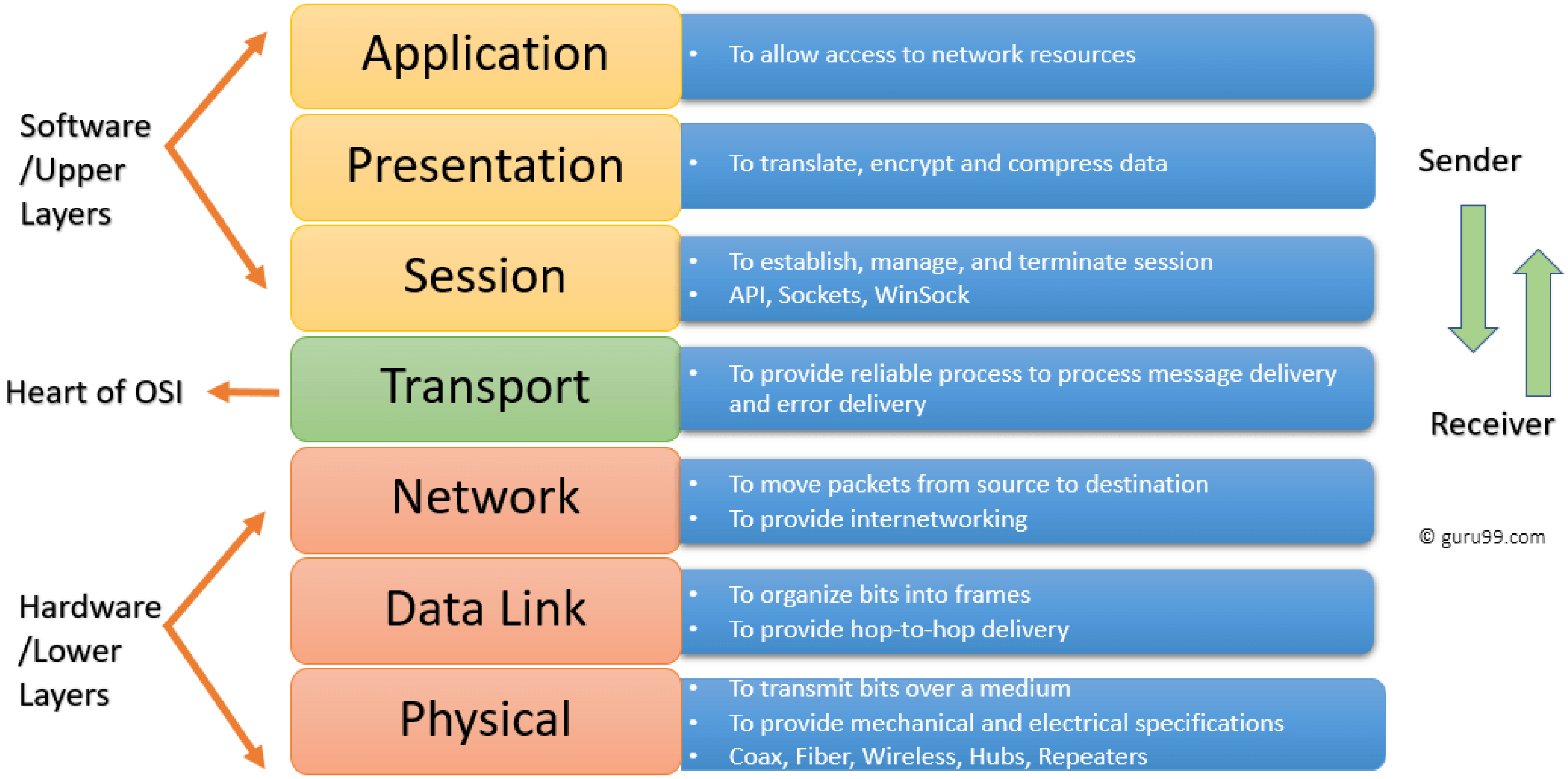
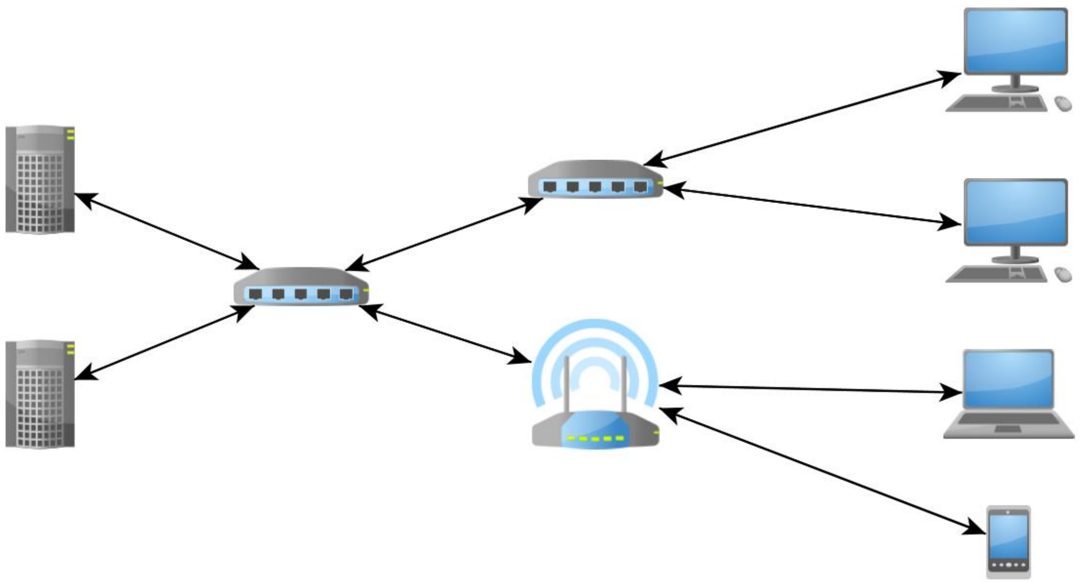
OSI COMMUNICATION MODEL



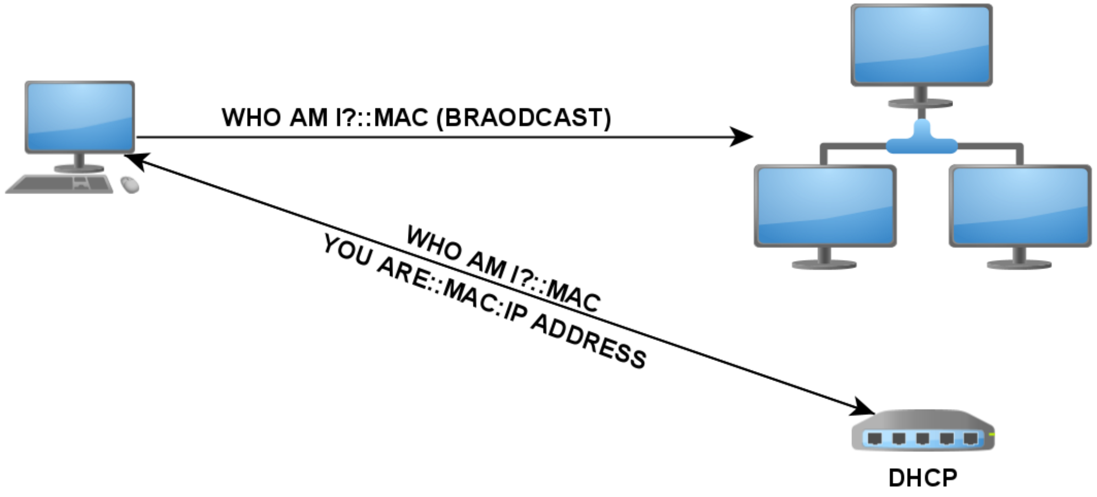
OSI COMMUNICATION MODEL

- Physical layer:
  - Main technologies: wired and wireless
  - Wired assumes ethernet or optical cables and router
  - Wireless uses radio signals to transfer data
  - Access point in wireless technology is identified by unique SSID.



OSI COMMUNICATION MODEL

- Data Link layer:
  - ARP (Address resolution protocol) is working on data link layer
  - Every network interface has its unique MAC (Media Access Control) address
  - ARP is connecting (mapping) IP address (which belongs to Network layer) to MAC address (which belongs to Data Link layer concept)



OSI COMMUNICATION MODEL

- Network layer:
  - **IP** is working on network layer.
  - Every device on computer network is uniquely addressed from within the network by assigning the IP address for the purpose of distinguishing from the other members of the same network.
  - Today there are two version of IP:
    - **IPv4**, 32-bit in format 8b.8b.8b.8b
      - Local IPv4: 128.0.0.0/1
    - **IPv6**, 128-bit in format 16b:16b:16b:16b:16b:16b:16b:16b:
      - Local IPv6: fe80:<converted MAC>
  - Local network segments:
    - 10.0.0.0 - 10.255.255.255
    - 172.16.0.0 - 172.31.255.255
    - 192.168.0.0-192.168.255.255

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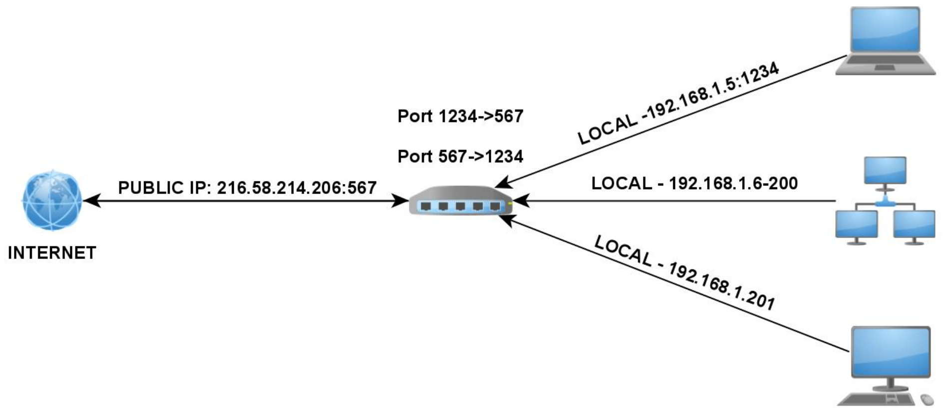
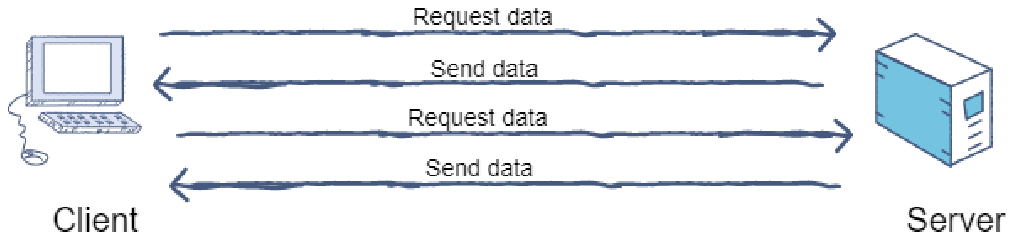
- Network layer:
  - NAT (Network Address Translation)Translates private IP address used within the LAN into public IP address used on the Internet and the other way around.
  - NAT uses pairs private_ip-private_port and public_ip-public_port to connect packets with corresponding destinations by storing the data inside NAT table.

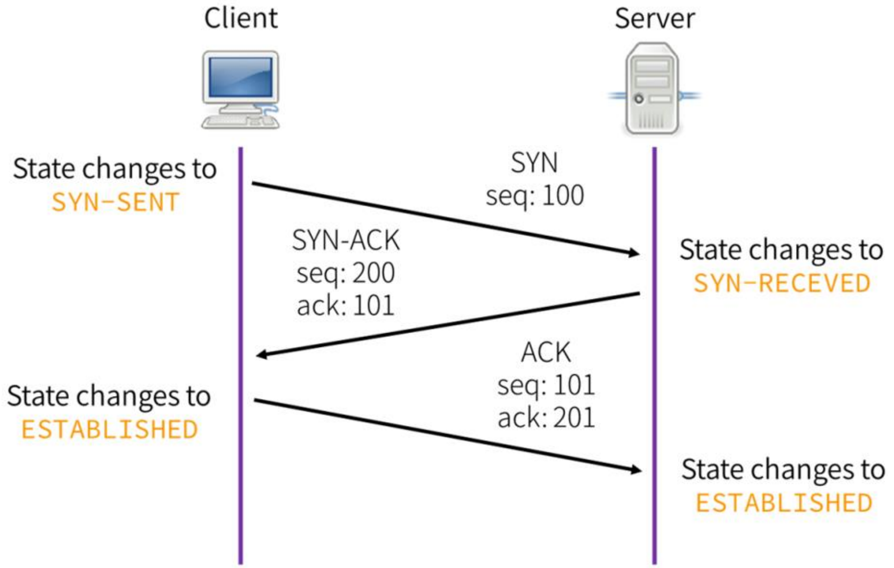
Private IP	Private Port	Public IP	Public Port	Destination IP	Destination Port
192.168.1.10	12345	202.203.111.2	67895	8.8.8.8	53
192.168.1.11	12346	202.203.111.2	67896	8.8.8.8	53



OSI COMMUNICATION MODEL

- Transport layer:
  - Sharing packets to different services is achieved by introducing the concept of PORTs
  - Two main protocols on Transport layer are TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).





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- Session layer:
  - Session is the interval between opening and closing the connection.
  - The responsibility of this layer is to establish, maintain and terminate sessions.
  - The concept of **checkpoints** is introduced so if communication fails, these points allow the session to resume from the last checkpoint, reducing the need to resend all data.
- Presentation layer:
  - Primary function of this layer is to act as a translator between the application and the network by ensuring that data is properly formatted and presented in a way that both the sender and receiver can understand.
  - Data Compression: Reduces the size of the data to be transmitted, ensuring that the data is efficiently sent over the network.
  - Data Encryption/Decryption: Handles the encryption of data before it is transmitted and decrypts it when received, ensuring data security and privacy during transmission (SSL and TLS operate at this layer).
  - Data formatting and character encoding: Ensures that data from the application layer is formatted appropriately for network transmission and deals with converting character sets between different systems (e.g., UTF-8, ASCII).

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- Application layer:
  - Responsible for providing interface to user.
  - Web browsers are working on this layer.
  - The main protocol on this layer is HTTP (Hypertext Transfer Protocol) which enables the Web site traffic over the Internet.
  - Other protocols working on application layer are SMTP (Simple Mail Transfer Protocol), FTP (File Transfer Protocol), IMAP (Internet Message Access Protocol), POP3 (Post Office Protocol), DNS (Domain Name System).

HTTP Protocol

- HTTP is used by web browsers to get the web page from the web server.
- The flow of HTTP communication includes sending a HTTP request by the client and receiving the HTTP response which is sent from the server.
- HTTP standard requires that every request has following structure:
  - HTTP method
  - Document URL
  - HTTP version
  - Request Header
  - Request Body (optional)

```
GET /primer.html HTTP/1.1
User-Agent: Mozilla/4.0 (compatible; MSIE5.01; Windows NT)
Host: www.singidunum.ac.rs
Accept-Language: en-us
Accept-Encoding: gzip, deflate
Connection: Keep-Alive
```